

Problem Set 9

Problem 1. [10 points]

(a) [5 pts] Show that of any $n + 1$ distinct numbers chosen from the set $\{1, 2, \dots, 2n\}$, at least 2 must be relatively prime. (*Hint:* $\gcd(k, k + 1) = 1$.)

(b) [5 pts] Show that any finite connected undirected graph with $n \geq 2$ vertices must have 2 vertices with the same degree.

Problem 2. [10 points] Under Siege!

Fearing retribution for the many long hours his students spent completing problem sets, Prof. Leighton decides to convert his office into a reinforced bunker. His only remaining task is to set the 10-digit numeric password on his door. Knowing the students are a clever bunch, he is not going to pick any passwords containing the forbidden consecutive sequences "18062", "6042" or "35876" (his MIT extension).

How many 10-digit passwords can he pick that don't contain forbidden sequences if each number 0, 1, ..., 9 can only be chosen once (i.e. without replacement)?

Problem 3. [50 points] Be sure to show your work to receive full credit. In this problem we assume a standard card deck of 52 cards.

(a) [4 pts] How many 5-card hands have a single pair and no 3-of-a-kind or 4-of-a-kind?

(b) [4 pts] For fixed positive integers n and k , how many nonnegative integer solutions $x_0, x_1, \dots, x_k$ are there to the following equation?

$$\sum_{i=0}^k x_i = n$$

(c) [4 pts] For fixed positive integers n and k , how many nonnegative integer solutions $x_0, x_1, \dots, x_k$ are there to the following equation?

$$\sum_{i=0}^k x_i \leq n$$

① ~~A~~ Any set consisting of $n+1$ elements will have at least two adjacent elements by the pigeonhole principle. Since the $\gcd(k, k+1)$ for any $k \in \mathbb{Z}^+$, any set of $n+1$ elements must have at least two coprime members. $\square$

② ~~B~~ Suppose, for sake of contradiction, that any finite undirected connected graph having $n \geq 2$ vertices does not have 2 vertices with the same degree. This means every vertex has a unique degree. But for such a graph having $n \geq 2$ vertices, the maximum degree of any vertex is $n-1$, which immediately leads to a contradiction because there are not enough unique values $\{1, \dots, n-1\}$ to assign to n vertices. Thus, any undirected connected graph having $n \geq 2$ vertices must have 2 vertices with the same degree. $\square$

② Using complementary counting & PIE:

almost $10!$ permutations in total

- $(6 \cdot 5!)$ permutations having substring "18062"

- $(7 \cdot 6!)$ permutations having substring "6042"

- $(6 \cdot 5!)$ permutations having substring "35876"

+ 0 permutations having substring "18062" and substring "6042"

+ 0 permutations having substring "18062" and substring "35876"

+ ~~$(2 \cdot 1!)$~~ permutations having substring "6042" having substring "35876"

$3 \cdot 2!$

double mistake

- 0 permutations having substring "18062" and substring "6042" and substring "35876"

~~$10! - 7! - 2(6!) + 2!$~~

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③ ~~14~~ Counting hands with a single pair and no 3-of-a-kind or 4-of-a-kind

$$\boxed{\binom{4}{2} \binom{13}{11} \cdot \binom{4}{1} \binom{12}{1} \cdot \binom{4}{1} \binom{11}{1} \cdot \binom{4}{1} \binom{10}{1}}$$

3!

⑤ For sequence $x_0, \dots, x_k$ such that $\sum_{i=0}^k x_i = n$, we are decomposing n into a sum consisting of $k+1$ integers. This is isomorphic to picking n scoops of ice cream from $k+1$ flavors, which is a combination with replacement problem:

$$\boxed{\frac{(n+k)!}{n! k!} = \binom{n+k}{n} = \binom{n+k}{k}}$$

⑥ From above, we can explicitly express the number of sequences $x_0, \dots, x_k$ such that $\sum_{i=0}^k x_i \leq n$:

$$\sum_{m=0}^n \binom{m+k}{m} = \binom{k}{0} + \binom{1+k}{1} + \binom{2+k}{2} + \dots + \binom{n+k}{n}$$

(CONT. ③E)

We know that the sequences that sum to some $m \leq n$ is disjoint from the set of sequences that sum to $l \leq n$ where $l \neq m$.

$$\sum_{m=0}^n \binom{m+k}{m} = \boxed{\binom{n+1+k}{n}} = \binom{n+(k+1)}{k+1}$$

(by the "hockey-stick" identity in Pascal's triangle)

③E ✓ There are $\binom{n}{2}$ possible edges on a simple undirected graph with n vertices. Thus, $\boxed{2^{\binom{n}{2}}}$ counts the number of possible simple undirected graphs on n vertices because each possible edge either is in a given graph or not.

④ ✓ By similar reasoning above, the number of directed graphs on n vertices is

$$\boxed{2^{2\binom{n}{2} + n}}$$

- (d) [4 pts] How many simple undirected graphs are there with n vertices?
- (e) [4 pts] How many directed graphs are there with n vertices (self loops allowed)?
- (f) [4 pts] How many tournament graphs are there with n vertices?
- (g) [4 pts] How many acyclic tournament graphs are there with n vertices?
- (h) [4 pts] How many numbers are there that are in the range $[1..700]$ which are divisible by 2, 5 or 7?
- (i) [9 pts] In how many ways can you arrange n books on k bookshelf (assuming the order of books on a shelf matters?)
- (j) [9 pts] How about if there has to be at least 1 book at each bookshelf?

Problem 4. [15 points] Give a combinatorial proof of the following theorem:

$$n2^{n-1} = \sum_{k=1}^n k \binom{n}{k}$$

(Hint: Consider the set of all length- n sequences of 0's, 1's and a single *.)

Problem 5. [15 points] At a congressional hearing, there are $2n$ members present. Exactly n of them are Democrats and n of them are Republicans. The members want to select a smaller subcommittee of size n from within those present at the hearing. However, since the Democrats currently hold majority, they want there to be more Democrats than Republicans in the committee. In how many ways can you select such a committee? (Hint: Consider two cases: n odd and n even.)

$$\sum_{d=\lfloor \frac{n}{2} \rfloor + 1}^n \binom{n}{d} \binom{n}{n-d} = \sum_{d=\lfloor \frac{n}{2} \rfloor + 1}^n \binom{n}{d}^2$$

(2) cont.)

③ ☒ There exists a bijection between simple undirected graphs on n vertices and tournament graphs on n vertices because the choice of an edge existing in the former is in one-to-one correspondence with which direction a directed edge should point between vertices u, v for all $u, v \in V$, where $|V| = n$. $\boxed{2^{\binom{n}{2}}}$

☒ An acyclic tournament graph imposes a total ordering on its nodes, thus, there are $\boxed{n!}$ such graphs for acyclic tournament graphs on n nodes.

☒ We want $|S_{1,700}^{2,5,7}|$ where $S_{1,700}^{2,5,7}$ is the set of integers $\in [1, 700]$ divisible by 2, 5, or 7.

Notice that $S_{1,700}^{2,5,7} = S_{1,700}^2 \cup S_{1,700}^5 \cup S_{1,700}^7$.

Using the principle of Inclusion-Exclusion allows us to count the elements of $S_{1,700}^{2,5,7}$ directly.

(24), cont.)

$$|S_{1,700}^{2,5,7}| = |S_{1,700}^2| + |S_{1,700}^5| + |S_{1,700}^7|$$

$$- |S_{1,700}^2 \cap S_{1,700}^5| - |S_{1,700}^2 \cap S_{1,700}^7|$$

$$- |S_{1,700}^5 \cap S_{1,700}^7|$$

$$+ |S_{1,700}^2 \cap S_{1,700}^5 \cap S_{1,700}^7|$$

$$= 350 + 140 + 100 - 70 - 50 - 20 + 10$$

=

(24) For each permutation of n books, we can allot j_i books on shelf i such that $\sum_{i=1}^k j_i = n$. The number of unique sequences $j_1, j_2, \dots, j_n$ is given by $\binom{n+k-1}{k-1}$, and there are $n!$ permutations of books, giving

$$n! \binom{n+k-1}{k-1}$$

If there needs to be at least 1 book per shelf, we have $\binom{n-1}{k-1}$ ways to arrange the remaining $n-k$ books per permutation, hence $n! \binom{n-1}{k-1}$ gives us the desired count.

④ Proof: We will count the number of unique n -length strings consisting of $k-1$ 1's, 1 *, and $n-k-1$ 0's as the basis of a combinatorial proof of

$$n2^{n-1} = \sum_{k=1}^n k \binom{n}{k}$$

LHS: There are 2^{n-1} strings of $n-1$ 0's and 1's. A * can be placed at any of n positions in each of these strings, thus giving $n2^{n-1}$ unique desired strings.

RHS: Each set of desired strings sharing the same count of 1's and *'s (i.e., same k) is disjoint from the other such sets having different k . Therefore, we can, by case work, sum over all possible k . For any given k , there are $\binom{n}{k}$ ways to place the 0's and k choices to place the * (the 1's are completely determined by these choices).

Since LHS & RHS count the same objects so they must be equal. $\square$

⑤ forgot to do! 

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